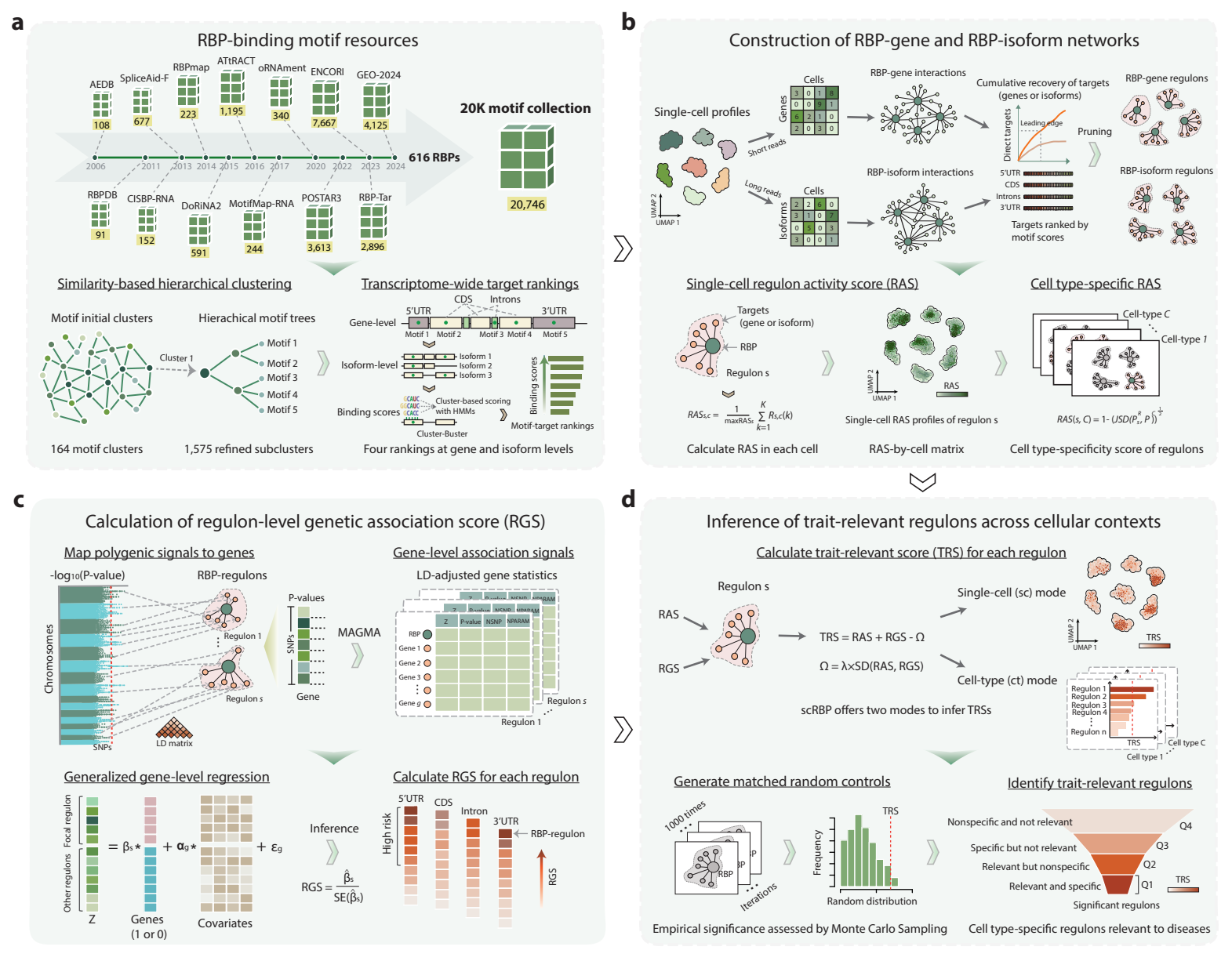
a
RBP-binding motif resources
AEDB
SpliceAid-F
RBPmap
ATtRACT
oRNAment
ENCORI
GEO-2024
108
677
223
1,195
340
7,667
4,125
2006
2011
2013
2014
2015
2016
2017
2020
2022
2023
2024
616 RBPs
20K motif collection
20,746
RBPDB
CISBP-RNA
DoRiNA2
MotifMap-RNA
POSTAR3
RBP-Tar
91
152
591
244
3,613
2,896
Similarity-based hierarchical clustering
Motif initial clusters
Hierachical motif trees
Cluster 1
Motif 1
Motif 2
Motif 3
Motif 4
Motif 5
164 motif clusters
1,575 refined subclusters
Transcriptome-wide target rankings
Gene-level
5'UTR
CDS
Introns
3'UTR
Motif 1
Motif 2
Motif 3
Motif 4
Motif 5
Isoform-level
Isoform 1
Isoform 2
Isoform 3
Binding scores
Cluster-based scoring with HMMs
Cluster-Buster
Binding scores
Motif-target rankings
Four rankings at gene and isoform levels
b
Construction of RBP-gene and RBP-isoform networks
Single-cell profiles
UMAP 2
UMAP 1
Short reads
Long reads
Genes
Isoforms
Cells
RBP-gene interactions
RBP-isoform interactions
Cumulative recovery of targets (genes or isoforms)
Direct targets
Leading edge
5'UTR
CDS
Introns
3'UTR
Targets ranked by motif scores
Pruning
RBP-gene regulons
RBP-isoform regulons
Single-cell regulon activity score (RAS)
Targets (gene or isoform)
RBP
Regulon s
$RAS_{s,c} = \frac{1}{\max RAS_s} \sum_{k=1}^{K} Rs,c(k)$
Calculate RAS in each cell
RAS
Single-cell RAS profiles of regulon s
RAS-by-cell matrix
Cell type-specific RAS
Cell-type C
Cell-type 1
$RAS(s, C) = 1 - (JSD(P_s^R, P))^{\frac{1}{2}}$
Cell type-specificity score of regulons
c
Calculation of regulon-level genetic association score (RGS)
Map polygenic signals to genes
-log10(P-value)
RBP-regulons
Chromosomes
Regulon 1
Regulon s
SNPs
LD matrix
P-values
SNPs
Gene
MAGMA
Gene-level association signals
LD-adjusted gene statistics
Z
P-value
NSNP
NPARAM
RBP
Gene 1
Gene 2
Gene 3
Gene g
Regulon 1
Regulon s
Generalized gene-level regression
Focal regulon
Other regulons
Z
$= \beta_s * + \alpha_g * + \varepsilon_g$
Genes (1 or 0)
Covariates
Inference
$RGS = \frac{\hat{\beta}_s}{SE(\hat{\beta}_s)}$
Calculate RGS for each regulon
High risk
5'UTR
CDS
Intron
3'UTR
RBP-regulon
RGS
d
Inference of trait-relevant regulons across cellular contexts
Calculate trait-relevant score (TRS) for each regulon
Regulon s
RAS
RGS
TRS = RAS + RGS - Ω
Ω = λ×SD(RAS, RGS)
Single-cell (sc) mode
Cell-type (ct) mode
UMAP 2
UMAP 1
TRS
Regulon 1
Regulon 2
Regulon 3
Regulon 4
Regulon n
TRS
Cell type C
Cell type 1
scRBP offers two modes to infer TRSs
Generate matched random controls
1000 times
RBP
Iterations
Frequency
TRS
Random distribution
Empirical significance assessed by Monte Carlo Sampling
Identify trait-relevant regulons
Nonspecific and not relevant
Specific but not relevant
Relevant but nonspecific
Relevant and specific
Q4
Q3
Q2
Q1
TRS
Significant regulons
Cell type-specific regulons relevant to diseases